

QUESTIONS 8 / 8 completed

Max-Min ▾

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.*;
2 public class Demo{
3     public static void main(String[] args){
4         Set<Integer>set=new HashSet<Integer>();
5
6         set.add(10);
7         set.add(20);
8         set.add(15);
9
10        System.out.println("Maximum:"+Collections.max(set));
11        System.out.println("Minimum:"+Collections.min(set));
12    }
13
14 }
```

OUTPUT

```
Maximum:20
Minimum:10
```

QUESTIONS 8 / 8 completed

LinkedListMap ▾

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.*;
2 public class Demo{
3     public static void main(String[] args){
4         LinkedHashMap<String,Integer>map=new LinkedHashMap<>();
5
6         map.put("apple",10);
7         map.put("Mango",30);
8         map.put("Grape",40);
9
10        for(String key:map.keySet()){
11            System.out.println(key);
12        }
13    }
14 }
```

OUTPUT

```
apple
Mango
Grape
```

QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Demo{
3     public static void main(String[] args){
4         PriorityQueue<Integer> pq = new PriorityQueue<Integer>();
5         pq.add(10);
6         pq.add(20);
7         pq.add(4);
8         pq.add(13);
9         System.out.println(pq);
10    }
11 }
```

OUTPUT

```
[4, 13, 10, 20]
```

QUESTIONS 8 / 8 completed

TreeMap ▾

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.*;
2 public class Demo{
3     public static void main(String[] args){
4         TreeMap<String,Integer> map= new TreeMap<>();
5         map.put("Apple",10);
6         map.put("Ball",20);
7         map.put("Cat",30);
8
9         for(Map.Entry<String,Integer>e:map.entrySet()){
10             System.out.println(e.getKey()+" "+e.getValue());
11         }
12     }
13 }
```

OUTPUT

```
Apple10
Ball20
Cat30
```

QUESTIONS 8 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.HashSet;
2 public class 192524453RC{
3     public static void main(String[] args){
4         HashSet<String> set=new HashSet<String>();
5
6         set.add("Apple");
7         set.add("Banana");
8         set.add("Cherry");
9
10        set.remove("Banana");
11        System.out.println(set);
12    }
13 }
14
```

QUESTIONS 8 / 8 completed

Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class SortDemo{
3     public static void main(String[] args){
4         ArrayList<Integer>list=new ArrayList<Integer>();
5         list.add(5);
6         list.add(2);
7         list.add(8);
8         list.add(1);
9         Collections.sort(list);
10        System.out.println(list);
11    }
12 }
```

OUTPUT

```
[1, 2, 5, 8]
```

QUESTIONS 8 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of

PROGRAM EDITOR

Java

Run

```
1 interface Character{
2     void move();
3     void attack();
4     void takeDamage();
5 }
6 class Hero implements Character{
7     public void move(){
8         System.out.println("Hero moves");
9     }
10    public void attack(){
11        System.out.println("Hero attacks");
12    }
13    public void takeDamage(){
14        System.out.println("Hero takes damage");
15    }
16    public static void main(String[] args){
17        Hero h=new Hero();
18        h.move();
19        h.attack();
20        h.takeDamage();
21    }
```

OUTPUT

```
Hero moves
Hero attacks
Hero takes damage
```

Logger

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class called "Log4jLogger"

```
1 interface Logger{
2     void debug(String message);
3     void info(String message);
4     void warning(String message);
5     void error(String message);
6 }
7 class ConsoleLogger implements Logger{
8     public void debug(String message){
9         System.out.println("[DEBUG]+"message);
10    }
11    public void info(String message){
12        System.out.println("[INFO]+"message);
13    }
14    public void warning(String message){
15        System.out.println("[WARNING]+"message);
16    }
17    public void error(String message){
18        System.out.println("[ERROR]+"message);
19    }
20 }
```

```
20     public class Main{
21         public static void main(String[]args){
22             Logger logger=new ConsoleLogger();
23
24             logger.debug("Debug message");
25             logger.info("Information message");
26             logger.warning("Warning message");
27             logger.error("Error message");
28         }
29     }
30 }
31 }
```